INCH–POUND

FF-B-2844/08
15 FEBRUARY 1994

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR; SINGLE ROW, RADIAL, NON-FILLING SLOT, MEDIUM SERIES (MS - SERIES) (CHROMIUM ALLOY STEEL)

The General Services Administration has authorized the use of this specification by all Federal Agencies.

The complete requirements for procuring the bearings described herein shall consist of the document and the latest issue of FF-B-2844/GEN.

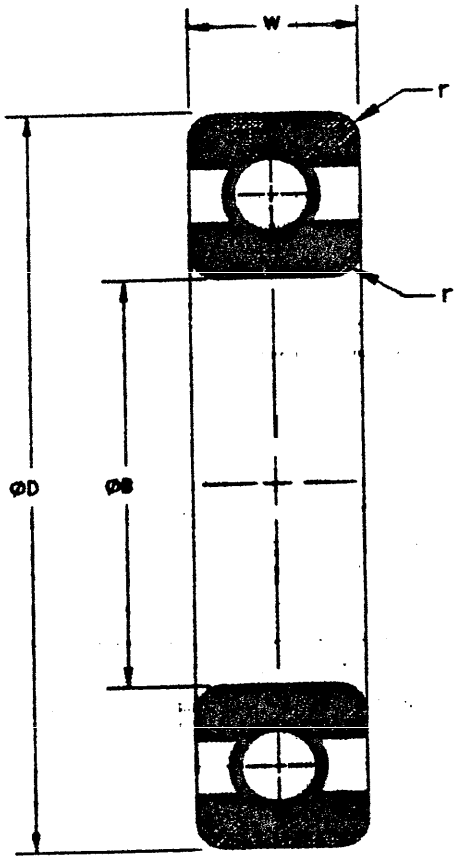


FIGURE 1.
OPEN(NO SHIELDS OR SEALS)

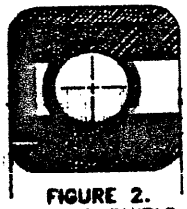


FIGURE 2.
SINGLE SHIELD

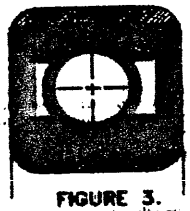


FIGURE 3.
DOUBLE SHIELD

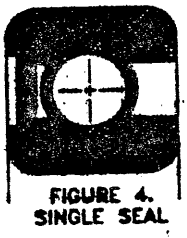


FIGURE 4.
SINGLE SEAL

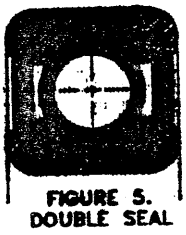


FIGURE 5.
DOUBLE SEAL



FIGURE 6.
SINGLE SHIELD AND SEAL

BEARING CONFIGURATIONS

FSC 3110

AMSC/NA

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REQUIREMENTS

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the bore codes in Table I.

2. MATERIAL

2.1 Rings and balls. The rings and balls of bearings covered by this specification shall be made of chromium-alloy steel 52100 (UNS 52986).

2.2 Closures. The shields and seals shall be fabricated from any material unless otherwise specified and shall be removable or non-removable type. Seals shall be impervious contact type. The number of shields and seals required shall be specified in the part number by using the codes listed in Table IV.

2.3 Cage material. Cages shall be manufactured from any material unless otherwise specified. The type of cage required shall be specified in the part number by using the codes listed in Table Manufactured.

3. TOLERANCE LIMITS

3.1 Radial internal clearance. The tolerance limits for radial internal clearance are specified in ANSI/AFBMA STD 20 and shall be specified in the part number by using the code listed in Table III.

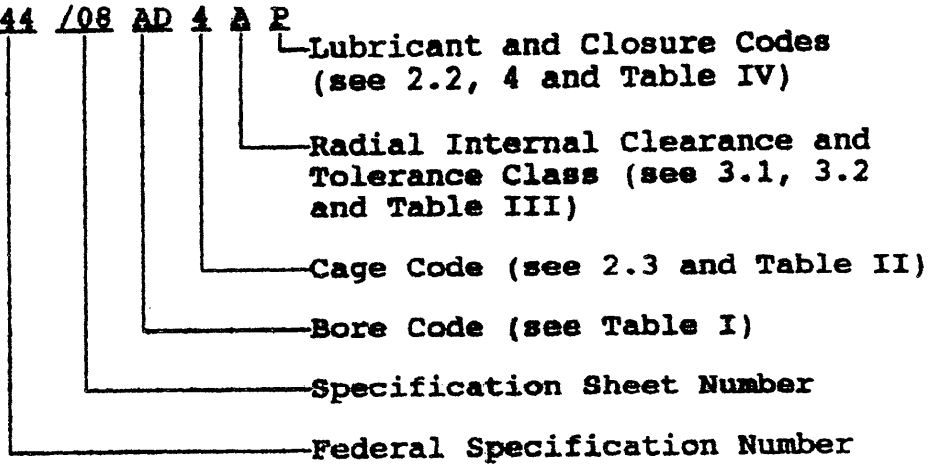
3.2 Tolerance class. The tolerance limits for tolerance class ABEC-1, and ABEC 5, shall be in accordance with ANSI/AFBMA STD 20 and shall be specified in the part number by using the code listed in Table III.

4 Lubricant or preservative compound The lubricant or preservative compound shall be specified in the part number by using the code listad in Table XV. Preservative compound shall be applied to all open bearings (no shields or seals) and to single closure bearings (single shield or single seal) that have no additional lubricant requirement. Thus, for these bearings, the bearing part number shall include the “C”, “F” or “L” suffix from table IV. For double closure or single closure bearings that require lubricant, the bearing part number shall include the preferred suffix from Table IV as required. When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25% to 50% filled for non-instrument size bearings and 25% to 40% filled for instrument bearings.

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5. **Part number.** The part number consists of the Federal Specification prefix letters FFB, the basic number of the Federal Specification, the specification sheet number, bore code, and alpha or numeric codes for cage, radial internal clearance and tolerance class, lubricant and closures.

EXAMPLE: FFB2844 /08 AD 4 A P



FFB2844/08AD4AP indicates - Bearing Bore 0.8750 inches; outside diameter 2.2500 inches; width 0.6875 inches; cage type, machined non metallic; radial internal clearance, tight; tolerance class, ABBC 1; single shield and single seal; lubricant in accordance with DoD-G-24508.

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TABLE I. BOUNDARY DIMENSIONS 1/

Bore Code	OB Bore Diameter (inch)	OB Bore Diameter (MM)	OD Outside Diameter	W Width	Chamfer Min. 2/	Dia. Series 3/	Dynamic Load Rating (lbs)
AA	0.5000	12.7000	1.6250	0.6250	.062	4	2465
AB	0.6250	15.8750	1.8125	0.6250	.062	3	2603
AC	0.7500	19.0500	2.0000	0.6875	.062	3	2748
AD	0.8750	22.2250	2.2500	0.6875	.062	3	3273
BA	1.0000	25.4000	2.5000	0.7500	.093	3	4513
BB	1.1250	28.5750	2.8125	0.8125	.093	3	5748
BC	1.2500	31.7500	3.1250	0.8750	.093	3	7040
BD	1.3750	34.9250	3.5000	0.8750	.093	3	8588
BE	1.5000	38.1000	3.7500	0.9375	.093	3	8728
BF	1.6250	41.2750	4.0000	0.9375	.093	4	10,414
BG	1.7500	44.4500	4.2500	1.0625	.093	3	11,348
BH	1.8750	47.6250	4.5000	1.0625	.093	4	13,258
CA	2.0000	50.8000	4.5000	1.0625	.093	3	13,258
CB	2.2500	57.1500	5.0000	1.2500	.125	3	15,365
CC	2.5000	63.5000	5.5000	1.2500	.125	3	16,731
CD	2.7500	69.8500	6.2500	1.3750	.125	3	21,632
DA	3.0000	76.2000	7.0000	1.5625	.125	3	24,326
DB	3.2500	82.5500	7.5000	1.5625	.156	3	26,113
DC	3.3750	85.7250	7.5000	1.5625	.156	3	26,108
DD	3.5000	88.9000	8.1250	1.7500	.156	3	28,716

1/ All dimensions are in inches unless otherwise specified.
2/ The chamfer on bearings will clear a maximum fillet radius equal to the minimum chamfer.
3/ Diameter series is a soft conversion of the inch outside diameter and width in comparison with the boundary dimension in ANSI/AFBMA STD 20.

TABLE II. CAGES

CODE	CAGE
1	Pressed Steel or Molded Plastic
2	Machined Brass or Bronze
3	Machined Non Metallic (phenolic)
4	Pressed Brass

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Table III. RADIAL INTERNAL CLEARANCE AND TOLERANCE CLASS

Code	Symbol	Tolerance
A B C D	2 (Tight) 0 (Normal) 3 (Loose) 4 (Extra Loose)	ABEC 1
E F G H	2 (Tight) 0 (Normal) 3 (Loose) 4 (Extra Loose)	ABEC 5

Table IV. LUBRICANTS AND CLOSURES

Code	Lubricant	Closure
A B C	DOD-G-24508 MIL-G-81322 Preservative 1/	OPEN (FIG 1)
D E F	DOD-G-24508 MIL-G-81322 Preservative 1/	SINGLE SHIELD (FIG 2)
G H	DOD-G-24508 MIL-G-81322	DOUBLE SHIELD (FIG 3)
J K L	DOD-G-24508 MIL-G-81322 Preservative 1/	SINGLE SEAL (FIG 4)
M N	DOD-G-24508 MIL-G-81322	DOUBLE SEAL (FIG 5)
P R	DOD-G-24508 MIL-G-81322	SINGLE SHIELD AND SEAL (FIG 6)

1/ Preservative shall be in accordance with MIL-P-197.

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